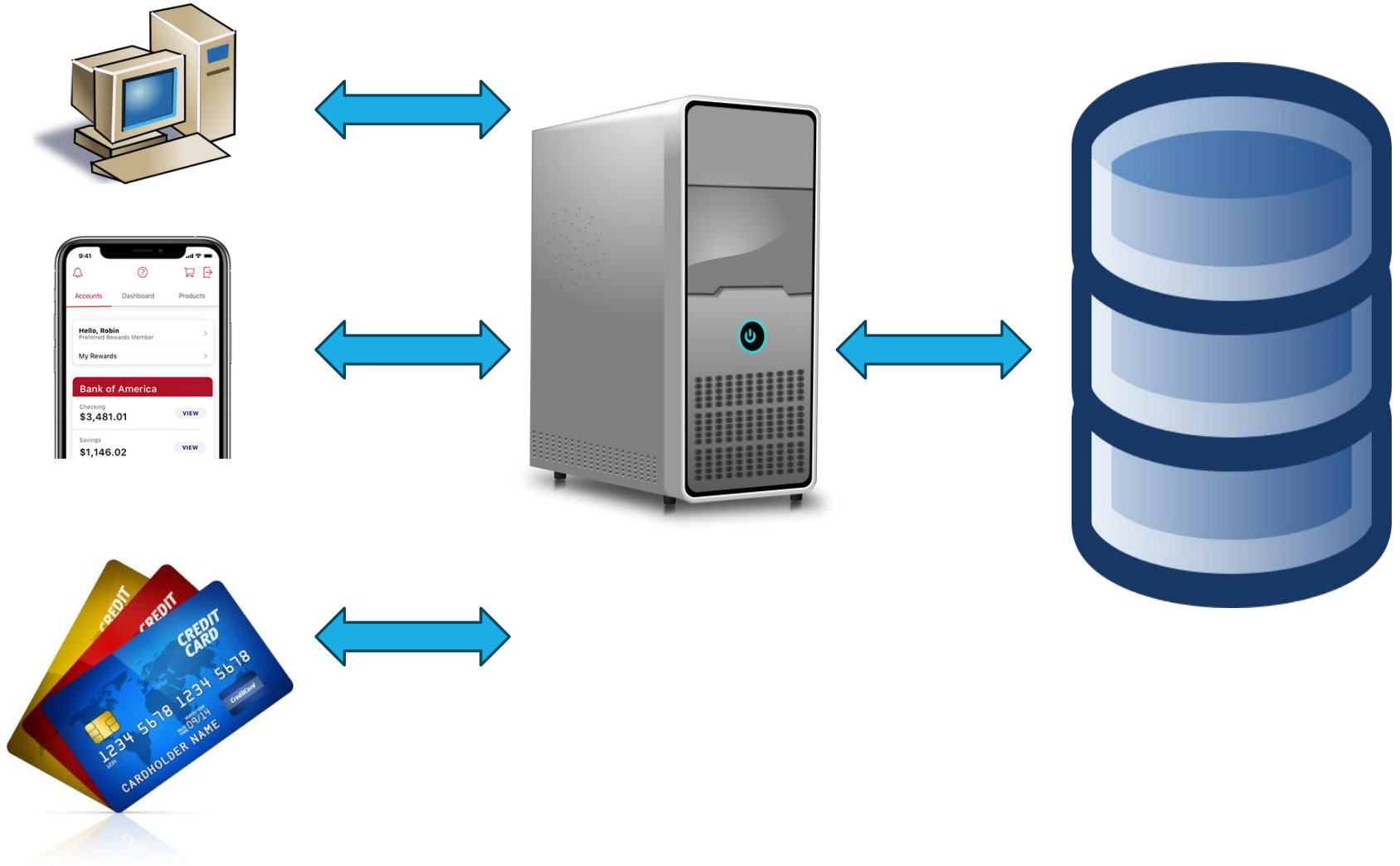


Web store (eCommerce)



Bank system



Home Remodeling



Business to Database

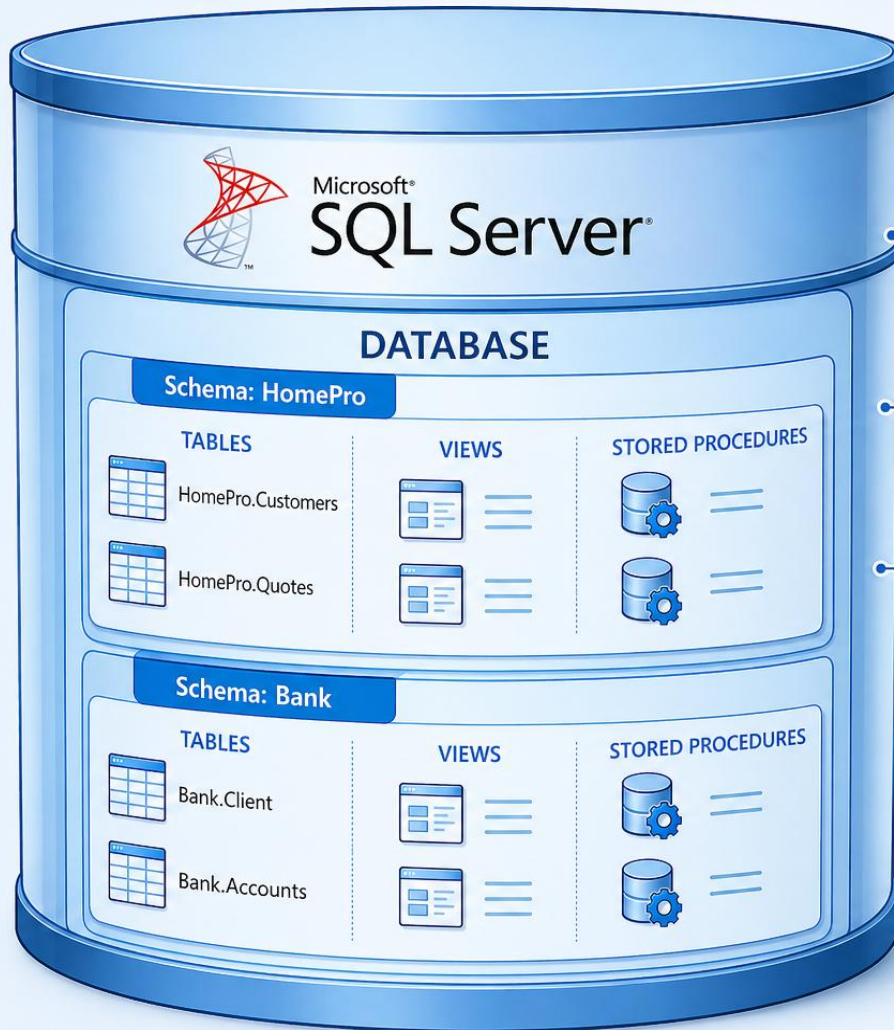
Bank

Business Entity	Database Table
Clients	Clients
Accounts	Accounts
Transactions	Transactions

Home remodeling (HomePro)

Business Entity	Database Table
Customers	Customers
Scheduled work	Schedules
Remodeling quotes	Quotes

Database



Microsoft SQL Server

A relational database management system that organizes data into databases.



DATABASE

A collection of related data and database objects.



SCHEMA

A logical container that groups database objects.



TABLE

Stores data in rows and columns.



VIEW

A virtual table based on the result of a SELECT query.



STORED PROCEDURE

A precompiled set of SQL statements that can be executed as a single unit.

Table

Table - collection of related data held in a structured format within a database. It consists of rows, and columns

- **Row, record** - represents a single piece of data or a unique record, business item
- **Column, field** - represents a specific attribute or field associated with the data

CustomerId	FirstName	LastName	Email	Phone	AltPhone	FaxNumber	ZipCode	NewsLetter
1	John	Smith	John@gmail.com	703-543-3302	703-543-3302	NULL	22201	1
2	Jeremy	Smith	Jeremy@gmail.com	723-543-3302	NULL	NULL	22203	0
3	Mark	Long	MarkLong@Yahoo.com	722-366-5588	NULL	NULL	22031	1
4	Bob	James	bob@microsoft.com	703-366-9632	NULL	703-455-9632	22221	0
5	Adam	Marcos	adam@Marcos.com	703-566-0000	NULL	703-366-0000	22001	1

Database design

Customers + Schedules relationship

CustomerId	FirstName	LastName	Email	Phone	ZipCode	ScheduleDescription	DateNeeded	JobType
1	John	Smith	John@gmail.com	703-543-3302	22201	Kitchen remodel needed	2013-10-10	Remodeling
2	Jeremy	Smith	Jeremy@gmail.com	723-543-3302	22203	Decorating help for dinig room	2013-10-15	Decorating
3	Mark	Long	MarkLong@Yahoo.com	722-366-5588	22031	Kitchen remodel needed	2015-11-29	Remodeling
3	Mark	Long	MarkLong@Yahoo.com	722-366-5588	22031	Garade rebuild	2016-12-31	Rebuild

Problems:

1. Duplicated data
2. Updated problem
3. Possible data ambiguity

Database design (Normalization)

Solution

Each table contains information about single functional item.

Primary key

Customers

CustomerId	FirstName	LastName	Email	Phone	ZipCode
1	John	Smith	John@gmail.com	703-543-3302	22201
2	Jeremy	Smith	Jeremy@gmail.com	723-543-3302	22203
3	Mark	Long	MarkLong@Yahoo.com	722-366-5588	22031

Relationship



Schedules

ID	CustomerId	Description	DateNeeded	Job Type
1	1	Kitchen remodel needed	2013-10-10	Remodeling
2	2	Decorations help for dinig room	2013-10-15	Decorating
3	3	Kitchen remodel needed	2015-11-29	Remodeling
4	3	Garade rebuild	2016-12-31	Rebuild

Foreign Key

Schema HomePro

HomePro.Customers

Primary Key

CustomerId	FirstName	LastName	Email	Phone	AltPhone	FaxNumber	ZipCode	NewsLetter	State	Age
1	John	Smith	John@gmail.com	703-543-3302	703-543-3302	NULL	22201	1	VA	18
2	Jeremy	Smith	Jeremy@gmail.com	723-543-3302	NULL	NULL	22203	0	NY	23
3	Mark	Long	Mark.Long@Yahoo.com	722-366-5588	NULL	NULL	22031	1	CA	64
4	Bob	James	bob@microsoft.com	703-366-9632	NULL	703-455-9632	22221	0	VA	37
5	Adam	Marcos	adam@Marcos.com	703-566-0000	NULL	703-366-0000	22001	1	NC	41

HomePro.Schedules

Primary Key

ID	CustomerId	Description	DateNeeded	Job Type
1	1	Kitchen remodel needed	2013-10-10	Remodeling
2	2	Decorating help for dinig room	2013-10-15	Decorating
3	3	Kitchen remodel needed	2015-11-29	Remodeling
4	3	Garade rebuild	2016-12-31	Rebuild

Foreign key

HomePro.Quotes

Primary Key

ID	CustomerId	Description	Estimation
1	1	Kitchen remodel	210.55
2	3	Quote with discount	875.55
3	3	Quote with additional work	10000.00

Foreign key

Schema Bank

Bank.Clients

Primary Key	ClientId	FirstName	LastName	Phone	Email	State	Age	Type
1	1	John	Smith	703-543-3302	John@gmail.com	VA	33	Private
2	2	Jeremy	Smith	723-543-3302	Jeremy@gmail.com	WA	19	Private
3	3	Long	Mark	722-366-5588	MarkLong@Yahoo.com	TN	41	Private
4	4	Bob	James	703-366-9632	bob@microsoft.com	VA	28	Business
5	5	Adam	Marcos	703-566-0000	adam@Marcos.com	CA	38	Business
6	6	Jason	Boley	345-234-9784	json@blabla.com	NY	31	Business
7	7	Tom	Soyer	572-223-5392	stom@hotmail.com	NJ	49	Private

Bank.Accounts

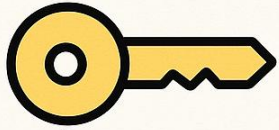
Primary Key	AccountNum	ClientId	Balance	Type
1	1	1	10200.00	CHECKING
2	2	1	3550.00	CREDIT
3	3	2	1001.00	CHECKING
4	4	2	150.00	CREDIT
5	5	3	1303.00	CHECKING
6	6	3	25000.00	SAVING
7	7	4	15731.00	CHECKING
8	8	4	31014.00	SAVING
9	9	5	1724.00	CHECKING
10	10	5	3043.00	CREDIT
11	11	5	79320.00	SAVING

Bank.Transactions

Primary Key	TransactionId	AccountNumFrom	AccountNumTo	Amount	TransactionTime	Status
1	1	1	2	150.00	2015-01-10 00:00:00.000	Pending
2	2	1	4	1000.00	2016-02-11 00:00:00.000	Committed
3	3	1	8	100.00	2016-04-01 00:00:00.000	Rejected
4	4	1	9	343.55	2017-01-18 00:00:00.000	Pending
5	5	2	9	36.70	2016-12-10 00:00:00.000	Committed
6	6	3	9	100.00	2016-12-12 00:00:00.000	Committed
7	7	5	9	1500.00	2015-01-10 00:00:00.000	Committed
8	8	5	10	1500.00	2016-06-13 00:00:00.000	Rejected
9	9	9	10	2300.00	2016-11-30 00:00:00.000	Committed
10	10	9	11	15000.00	2017-01-01 00:00:00.000	Committed



Primary key



ID	Name	Age
1	Alice	30
2	Bob	25
3	Charlie	35
4	Diana	28



PRIMARY KEY

- The column(s) that has completely unique data throughout the table
- The main role of a primary key in a data table is to maintain the internal integrity of a data table.
- Table can have only one primary key.

Foreign key

- The column that links one table to another table's primary key or unique constraint
- Table can have any number of foreign keys defined.

Structured Query Language (SQL)

SQL is a language used for creating, storing, fetching and updating of data and database objects in database.

SELECT

SELECT is used to retrieve rows selected from one or more tables.

Basic syntax:

```
SELECT <columns>  
FROM <Table>  
WHERE <condition>  
ORDER BY <columns>
```

```
select *  
from HomePro.Customers  
where LastName = 'Smith'  
order by LastName
```

Select (examples)

```
Select * from HomePro.Customers;
```

```
Select FirstName, LastName
```

```
From HomePro.Customers
```

```
Order by LastName;
```

Where (char, varchar)

Select * from HomePro.Customers

Where LastName = 'Smith'

Select * from HomePro.Customers

Where LastName like 'S%'

Select * from HomePro.Customers

Where LastName like '_m%'

Where (numbers)

1. Where Age = 10
2. Where Age > 10
3. Where Age > 10 and Age < 40
4. Where Age >= 10 and Age <= 40
5. Where Age between 10 and 40
6. Where Age in (10, 20, 30)

Where (date)

1. Where DateNedeed = '2015-11-29'
2. Where DateNedeed > '2014-12-30'
3. Where DateNedeed between '2015-12-01' and '2015-12-31'

NULL values

- NULL is an unknown and undefined value.
- Arithmetic operation with NULL in SQL will return a NULL.
- ~~Where Value = Null~~ -> Where Value is Null

Question: How much money do you have?

Answer 1. I have \$10 Meaning: \$10

Answer 2. I have no money Meaning : \$0

Answer 3. I don't tell you Meaning : Null

Query with NULL

Select *

from **HomePro.Customers**

where AltPhone is null;